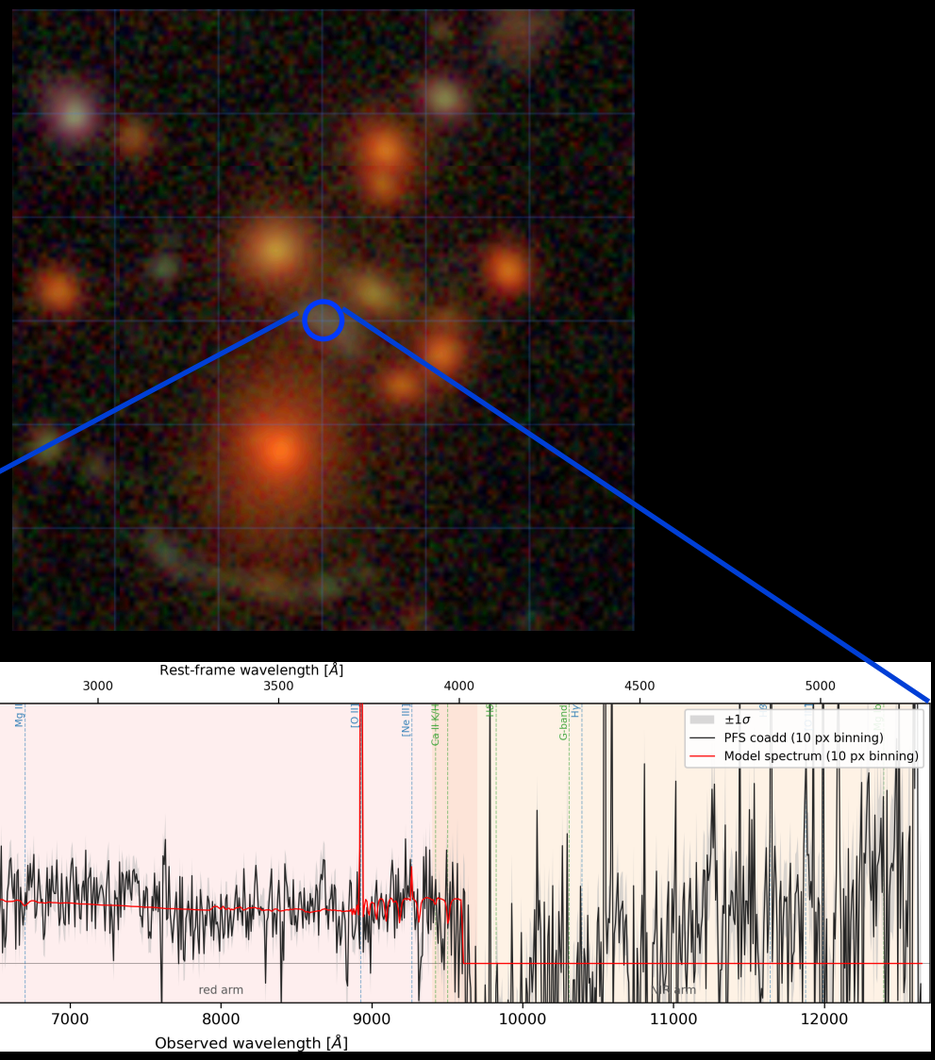
PFS Project 36

Spectroscopic Redshifts of Strong Gravitational Lenses in the SuGOHI Sample

Kenneth Wong (University of Tokyo)

PFS-AGN Internal Session

November 19, 2026



In collaboration with
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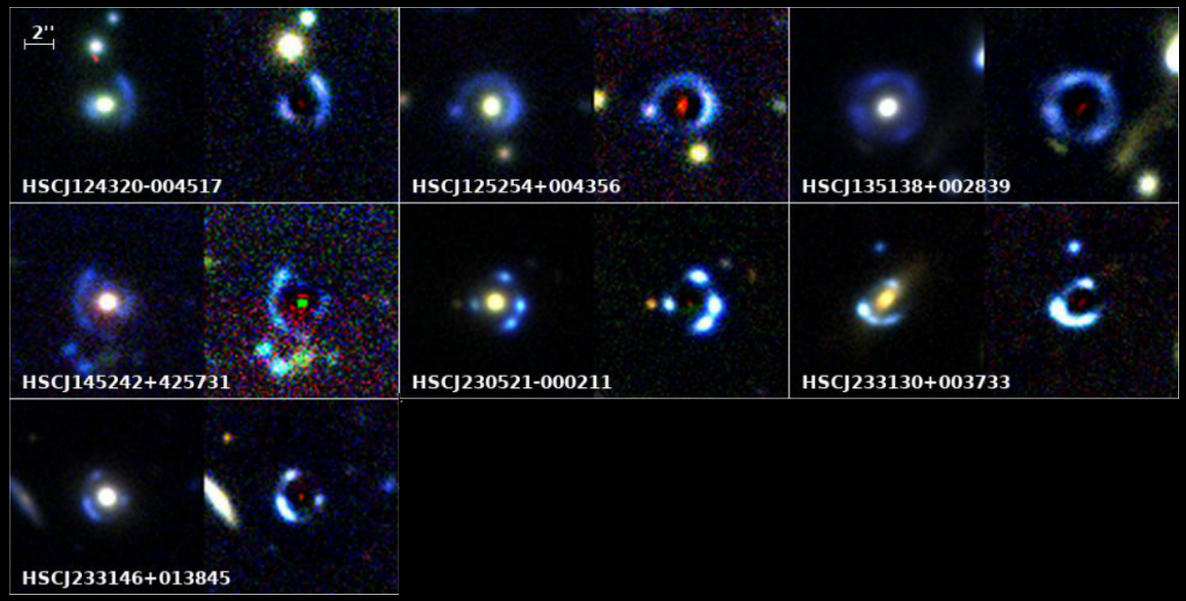
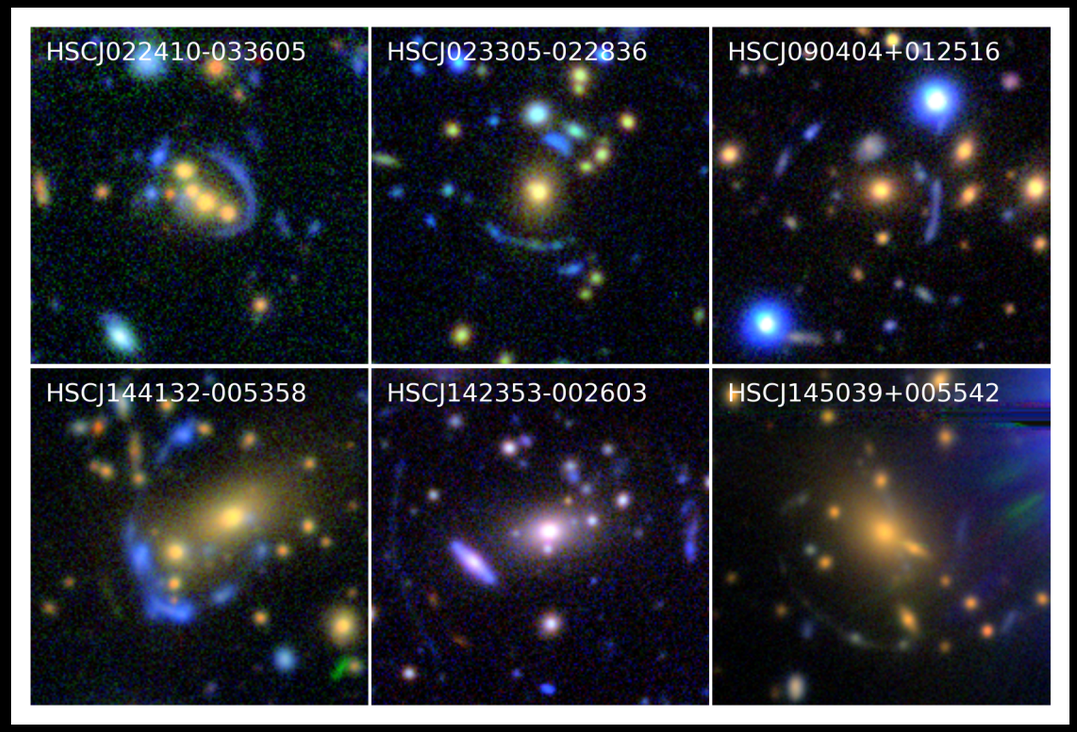
The SuGOHI Sample

- Subaru Hyper Suprime-Cam (HSC) SSP
  - >1000 deg^2 *grizy* imaging to z~26.2 (Wide)
  - ~0.6” median seeing (i-band)
- HSC SSP lens search program - Survey of Gravitational lens Objects in HSC Imaging (SuGOHI*), various methods include:
  - CHITAH (quasar-lens finder; Chan+2015)
  - YattaLens (arc finder; Sonnenfeld+2018)
  - Spectroscopic search method (Shu+2016)
  - Variability search method (Chao+2020)
  - Citizen science (Sonnenfeld+2020)
  - Machine learning techniques (Canameras+2021; Shu+2022; Jaelani+2024; Ishida+2025)
- To date, we have found:
  - ~2000 galaxy-scale lenses up to z_L~1
  - ~100 lensed quasar candidates
  - ~300 group/cluster scale lens candidates
- SuGOHI lenses tend to be at higher redshift ($z_L \gtrsim 0.5$) than other similar samples (e.g., DES), allowing us to study evolution of lens properties

*http://www-utap.phys.s.u-tokyo.ac.jp/~oguri/sugohi/



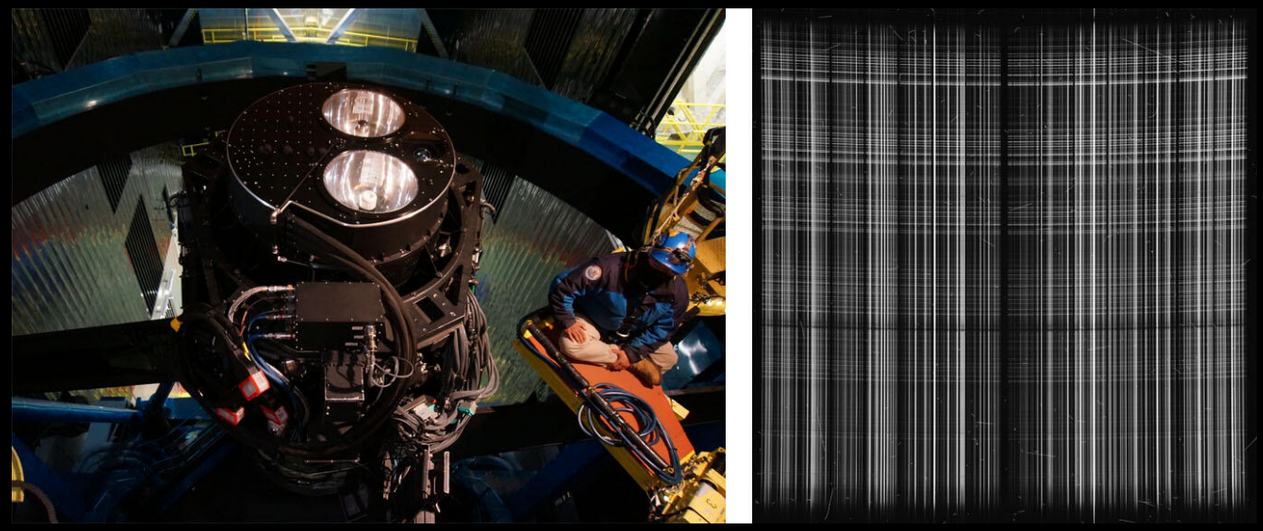
Wong+2018



Jaelani+2020

Measuring SuGOHI redshifts with PFS

- Most SuGOHI lenses lack source redshifts (z_S) and about ~half lack lens redshifts (z_L)
- Follow-up spectroscopy to target lenses one-by-one would take a lot of observing time
- CO-AGN working group has an approved ancillary target program (PI: Y. Toba)
  - includes ~2000 SuGOHI lens/source targets, prioritizing by candidate quality and source type
  - negligible fraction of total fibers
  - to date, 9 SuGOHI targets have been observed
- Estimate success rate of ~60% for source redshifts (depending on whether multiple emission lines fall within wavelength range)
- If completed, would be highly complementary to DESI redshifts (Shu & Li 2025; Li+2026) due to greater depth & wavelength coverage (i.e. higher redshifts)



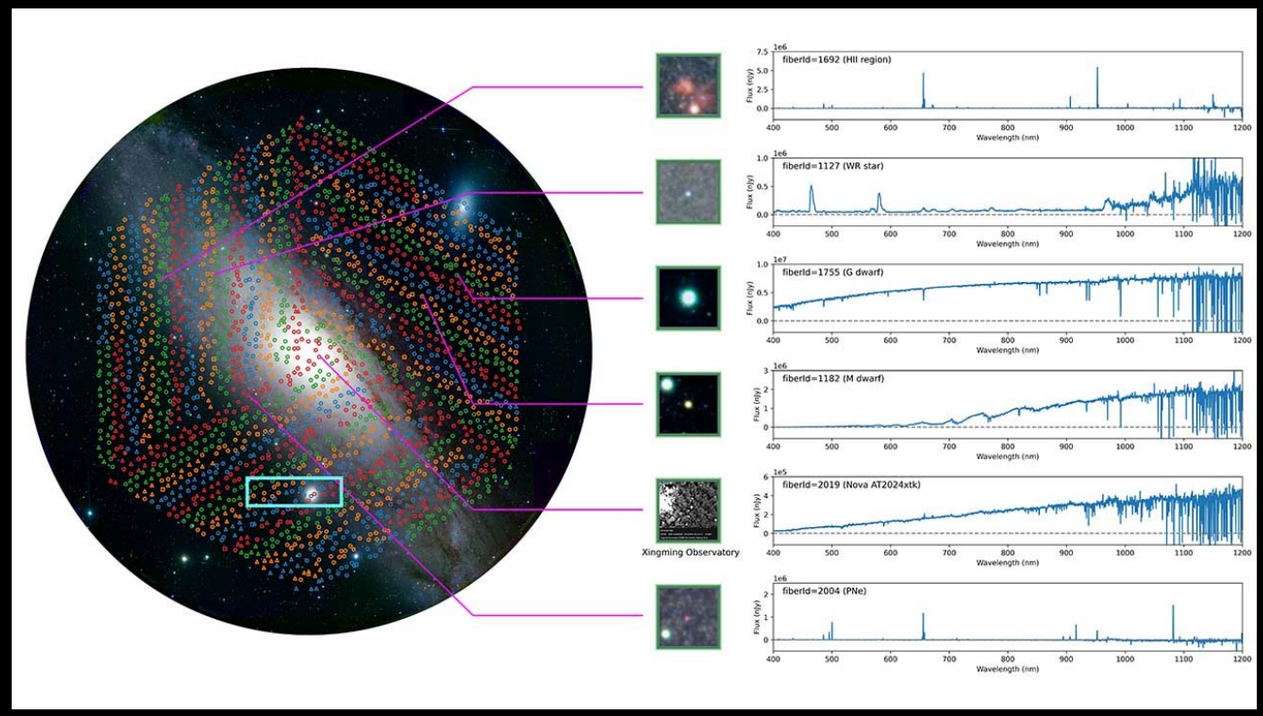
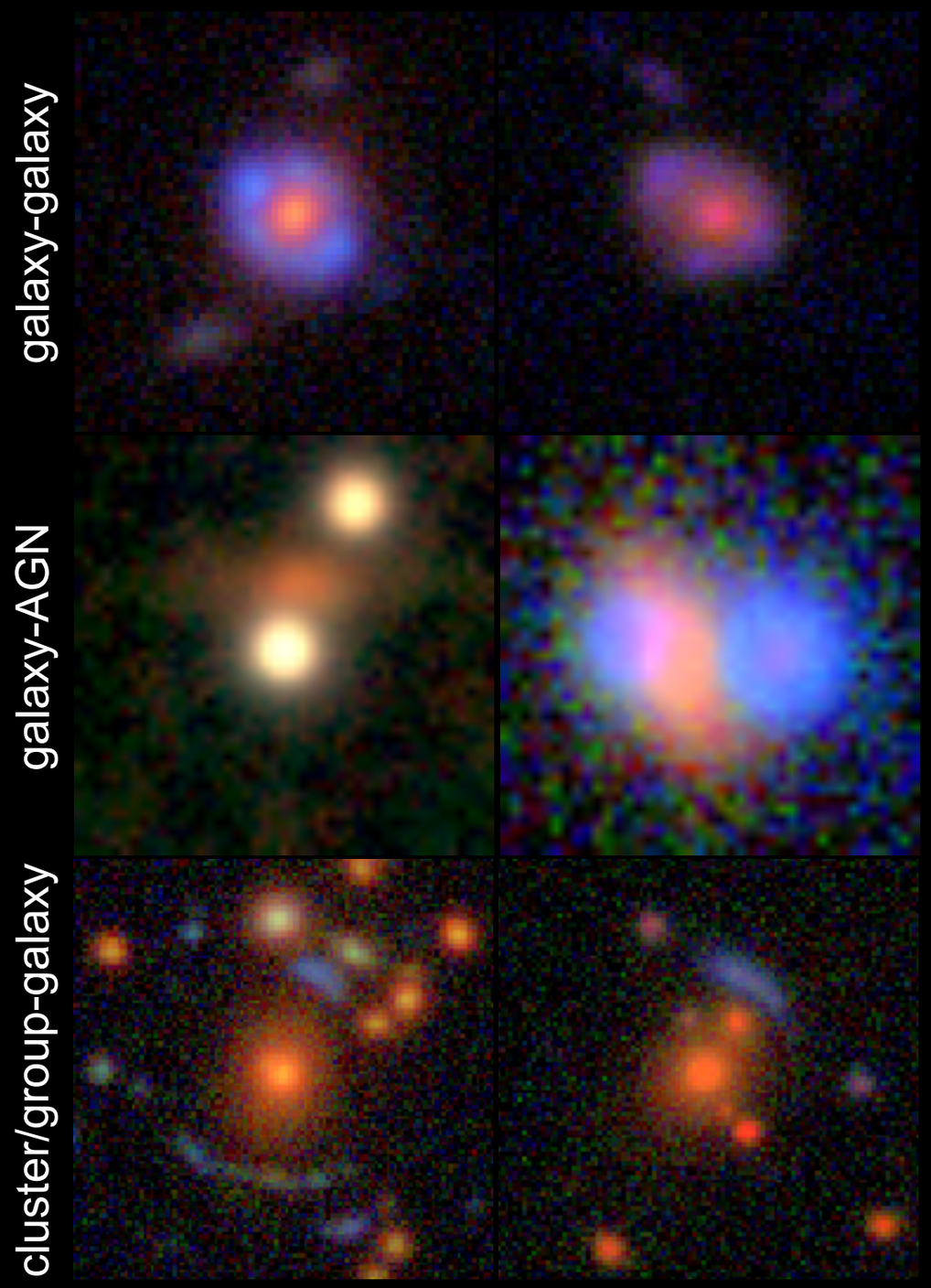


image credit: PFS project/IPMU/NAOJ

Target Selection

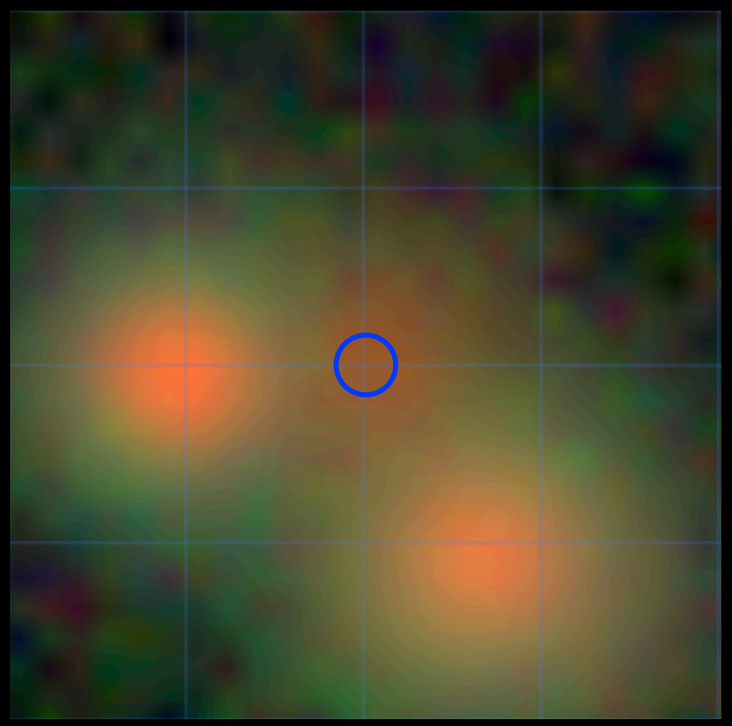
- SuGOHI lens candidates are given a grade based on visual inspection by multiple classifiers:
  - grade A: definite lenses ($P_{lens} \gtrsim 0.997$)
  - grade B: probable lenses ($P_{lens} \gtrsim 0.5$)
  - grade C: possible lenses ($P_{lens} \gtrsim 0.003$)
- Lens classification: galaxy-scale or cluster/group-scale
- Source classification: galaxy or quasar/AGN source
- Target priority
  1. grade A; missing z_L or z_S
  2. grade A; missing both
  3. grade B; missing z_L or z_S
  4. grade B; AGN source; missing both
  5. grade B; non-AGN source; missing both
- Exact fiber positions decided by visual inspection
  - when targeting source with multiple images, target brightest image
  - for extended image, target highest surface brightness region
- Final target list: $\gtrsim$ 2000 targets

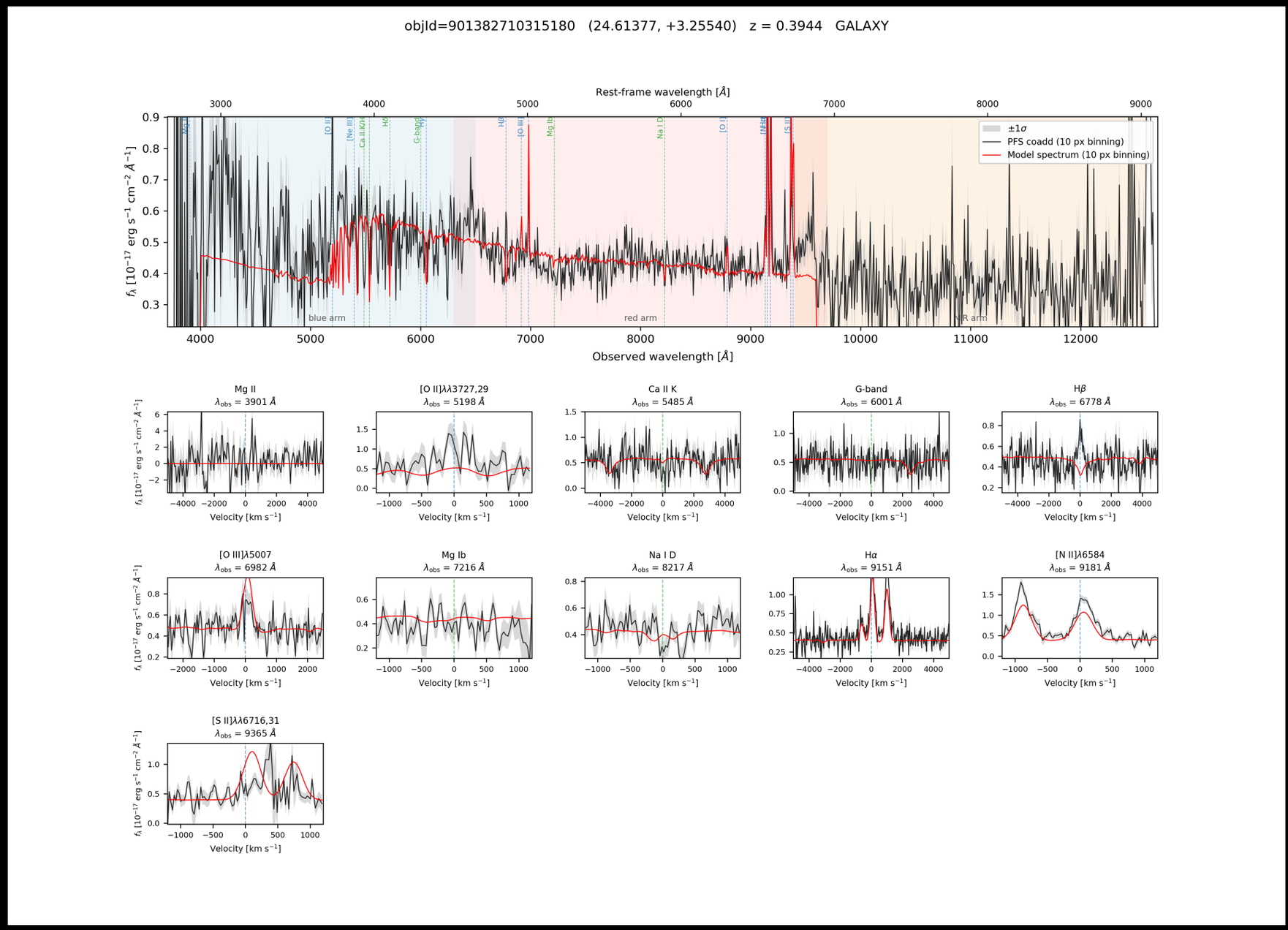
Examples of SuGOHI Lens Candidates



First PFS Spectra of SuGOHI Targets

HSC J013827+031518
z_L = ?
z_S = 2.406





successfully measured z_L

PFS pipeline redshift: z = 0.3944

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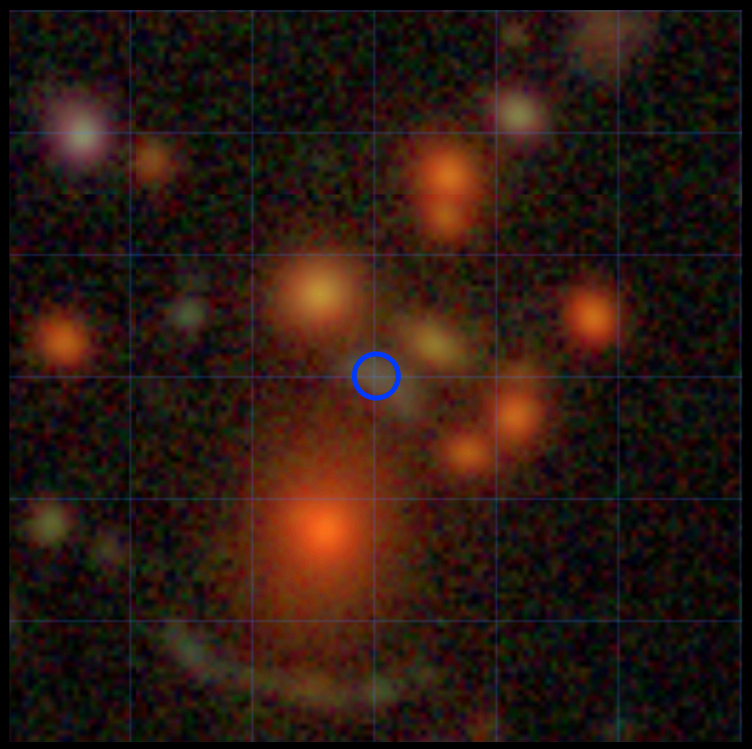
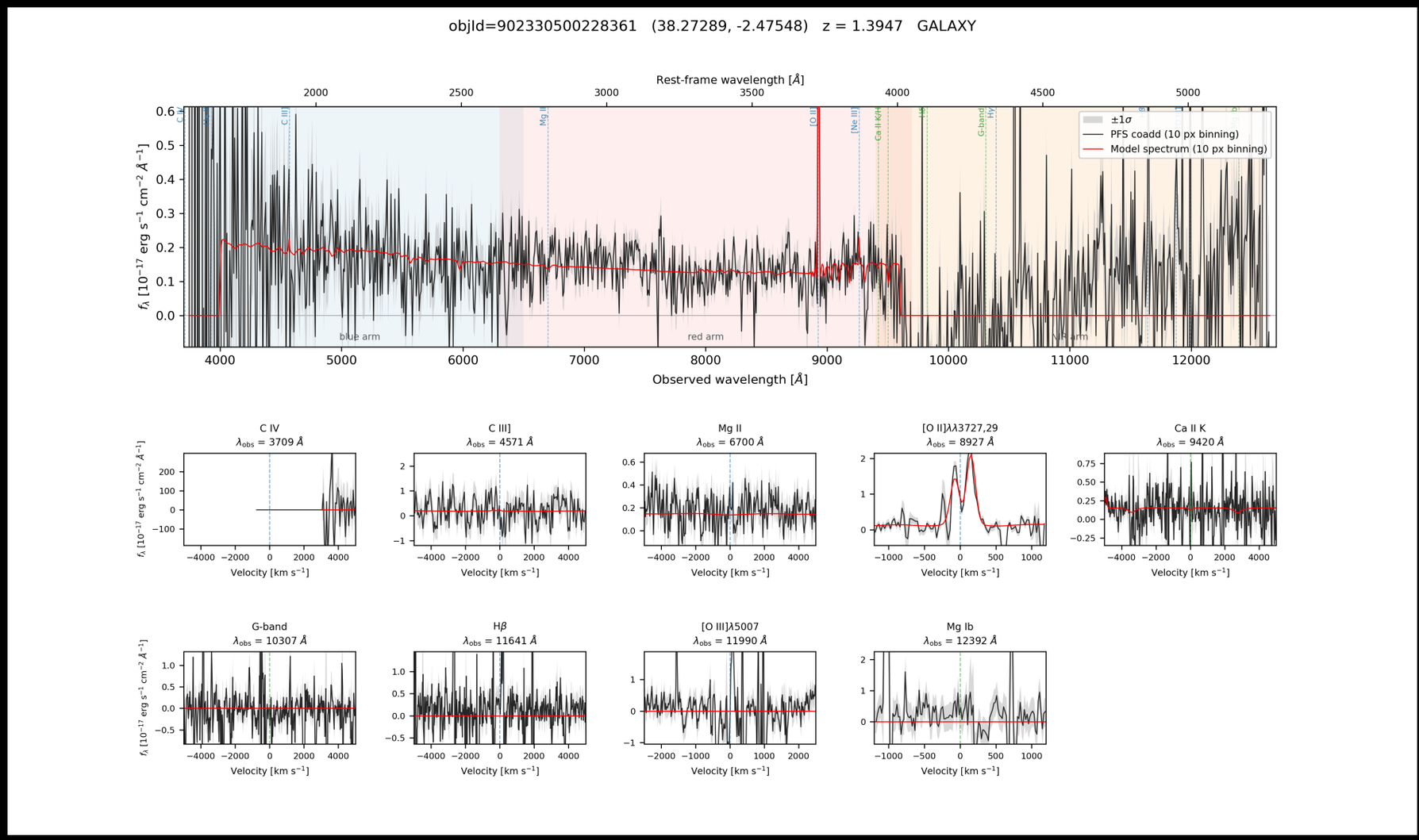
First PFS Spectra of SuGOHI Targets

HSC J023305-022836
z_L = 0.572
z_S = ?





successfully measured z_S

PFS pipeline redshift: z = 1.3947

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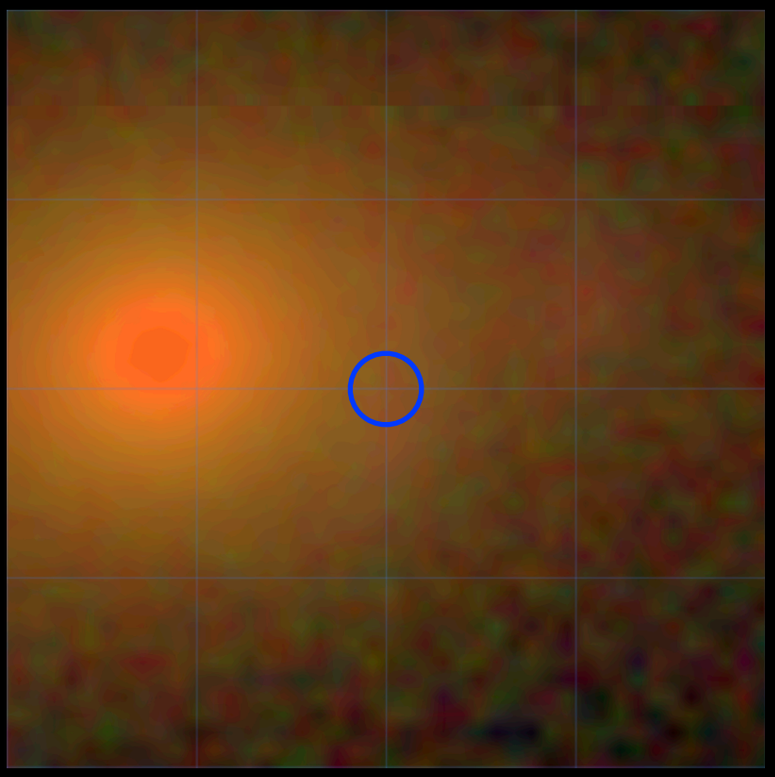
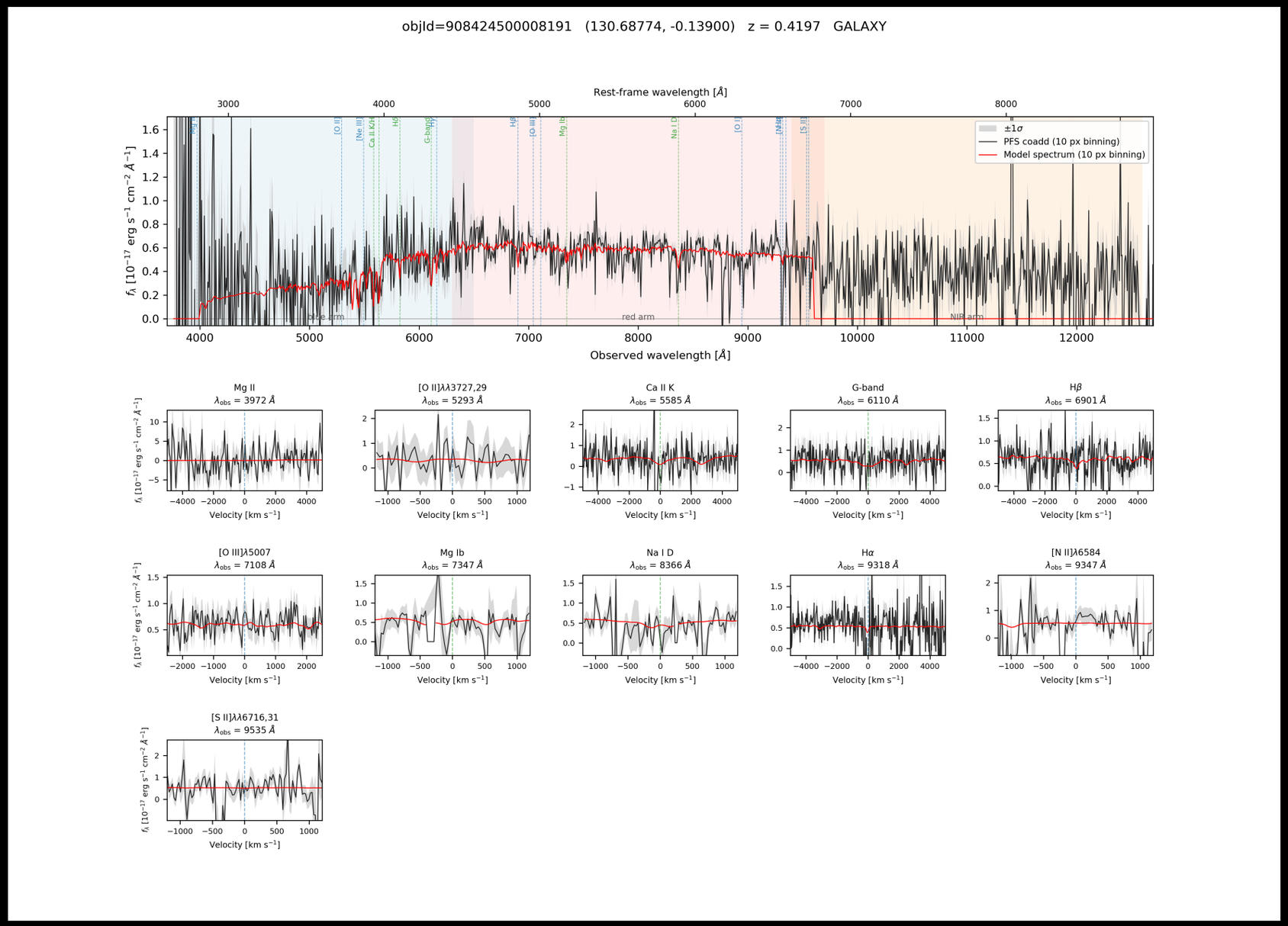
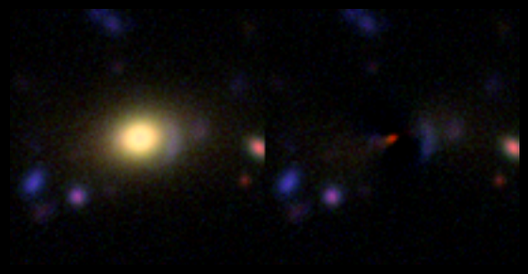
First PFS Spectra of SuGOHI Targets

HSC J084245-000819
z_L = 0.420
z_S = ?







unsuccessful - wanted to measure source redshift, measured lens redshift instead
alternative redshift solutions don't show clear fit
might need to perform our own fit?

PFS pipeline redshift: z = 0.4197

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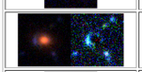
Future Plans

- Tentative plan
  - initial publication introducing the project and redshifts measured to date
  - publish redshifts as an extension of existing online SuGOHI catalog
  - update catalog as more redshifts are measured over the course of the PFS SSP

- Science enabled by PFS redshifts
  - systematic uniform lens modeling (e.g., Sonnenfeld+2019, Arwinata+ in prep.)
  - study of lens environments / line-of-sight structure (e.g., Wong+2018)
  - stellar/DM mass relation vs. redshift
  - correlation of lens parameters (e.g., mass, density slope, ellipticity) with redshift, lens environment
  - confirmation/follow-up of lensed quasars for time-delay cosmography

Questions/Issues

- Is there a way to force PFS pipeline to fit a specific template/redshift? how to handle blended spectra (i.e. simultaneously fit lens+source template)?
- Should there be a separate catalog for PFS redshifts, or is including it in SuGOHI catalog sufficient?
- What should be made public?
  - just redshifts (similar to most of the SuGOHI catalog)?
  - alternate redshift solutions?
  - ancillary measurements (e.g., velocity dispersion)?
  - best-fit template?

Cutout	Name	RA	Dec	z_{lens}	z_{source}	Lens type (GG: galaxy-galaxy; GQ: galaxy-quasar; CG: Cluster/group-galaxy; CQ: Cluster/group-quasar)	Grade (A: definite lens; B: probable lens; C: possible lens)	Reference
	HSCJ000008+022658	00:00:08.46	+02:26:58.0	-99.000	-99.000	GG		SuGOHI10
	HSCJ000018+001617	00:00:18.92	+00:16:17.7	-99.000	-99.000	GG	B	HOLISMOKES8
	HSCJ000020-002051	00:00:20.83	-00:20:51.0	-99.000	-99.000	GG	B	HOLISMOKES8 HOLISMOKES16
	HSCJ000051-002936	00:00:51.78	-00:29:36.6	-99.000	-99.000	GG		SuGOHI10
	HSCJ000106+010329	00:01:06.50	+01:03:29.8	-99.000	-99.000	GG	B	HOLISMOKES8 SuGOHI8
	HSCJ000114+001619	00:01:14.55	+00:16:19.7	0.664	1.027	GG	B	HOLISMOKES8 HOLISMOKES16
	HSCJ000326-002140	00:03:26.35	-00:21:40.8	-99.000	-99.000	GG	B	SuGOHI10
	HSCJ000327+021020	00:03:27.03	+02:10:20.9	-99.000	-99.000	GG	B	HOLISMOKES8 HOLISMOKES16 HOLISMOKES6+13

http://www-utap.phys.s.u-tokyo.ac.jp/~oguri/sugohi/

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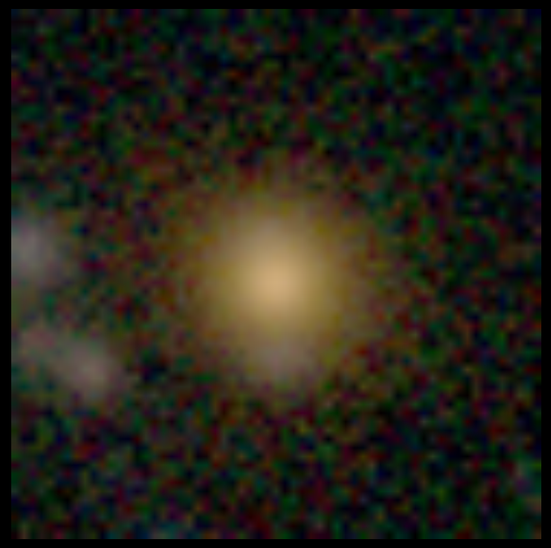
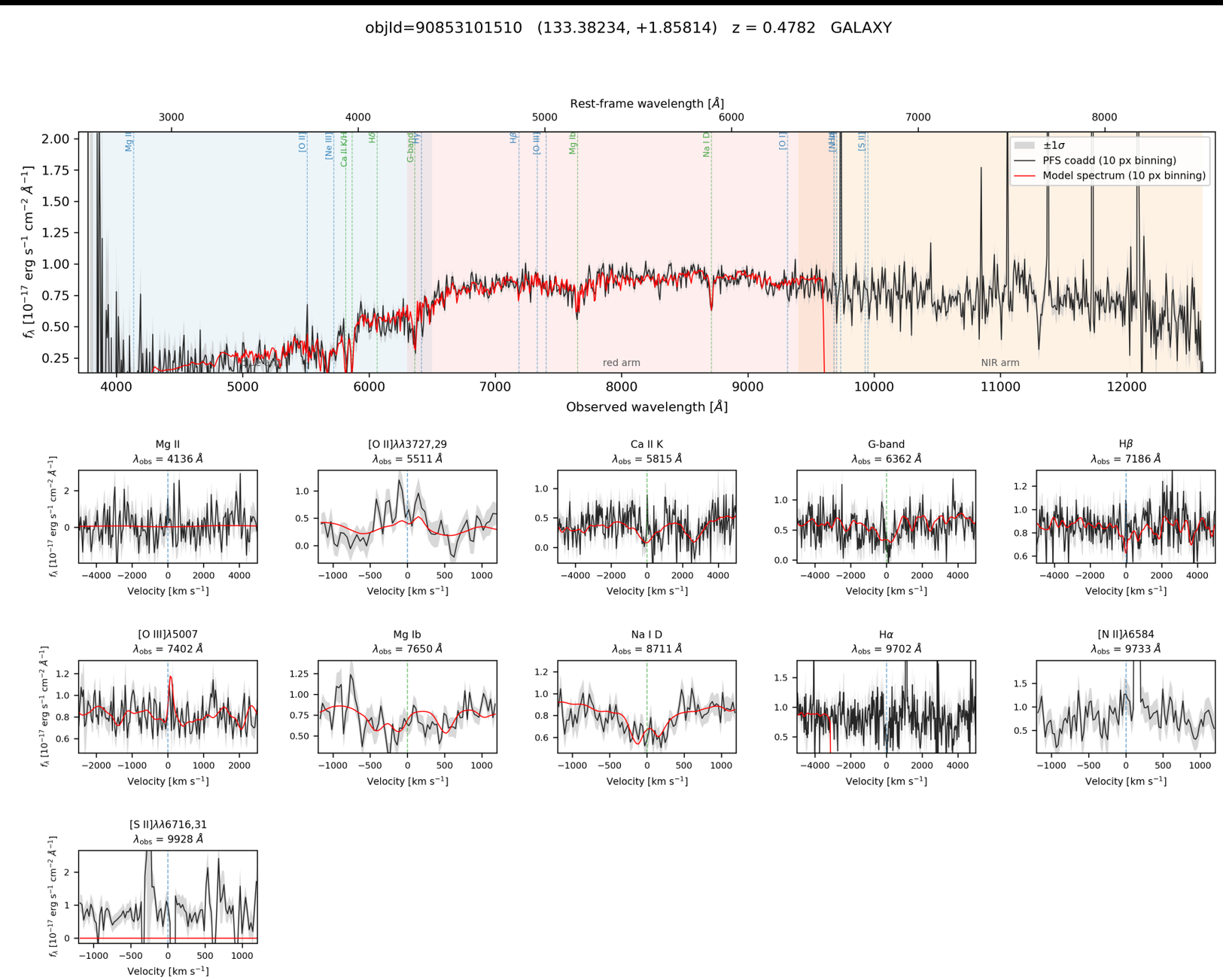
Backup slides





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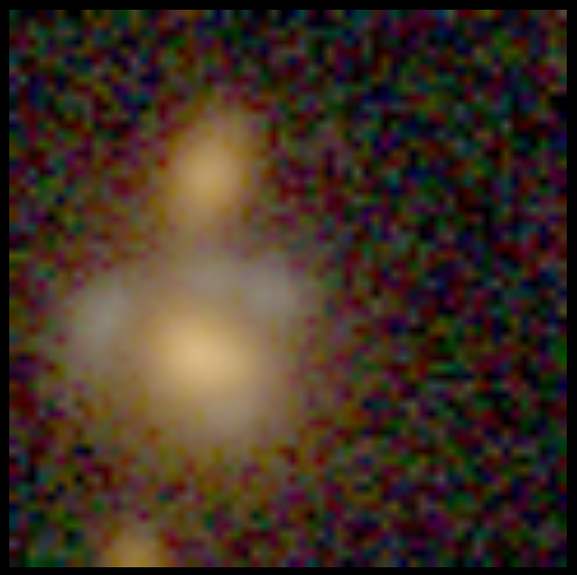

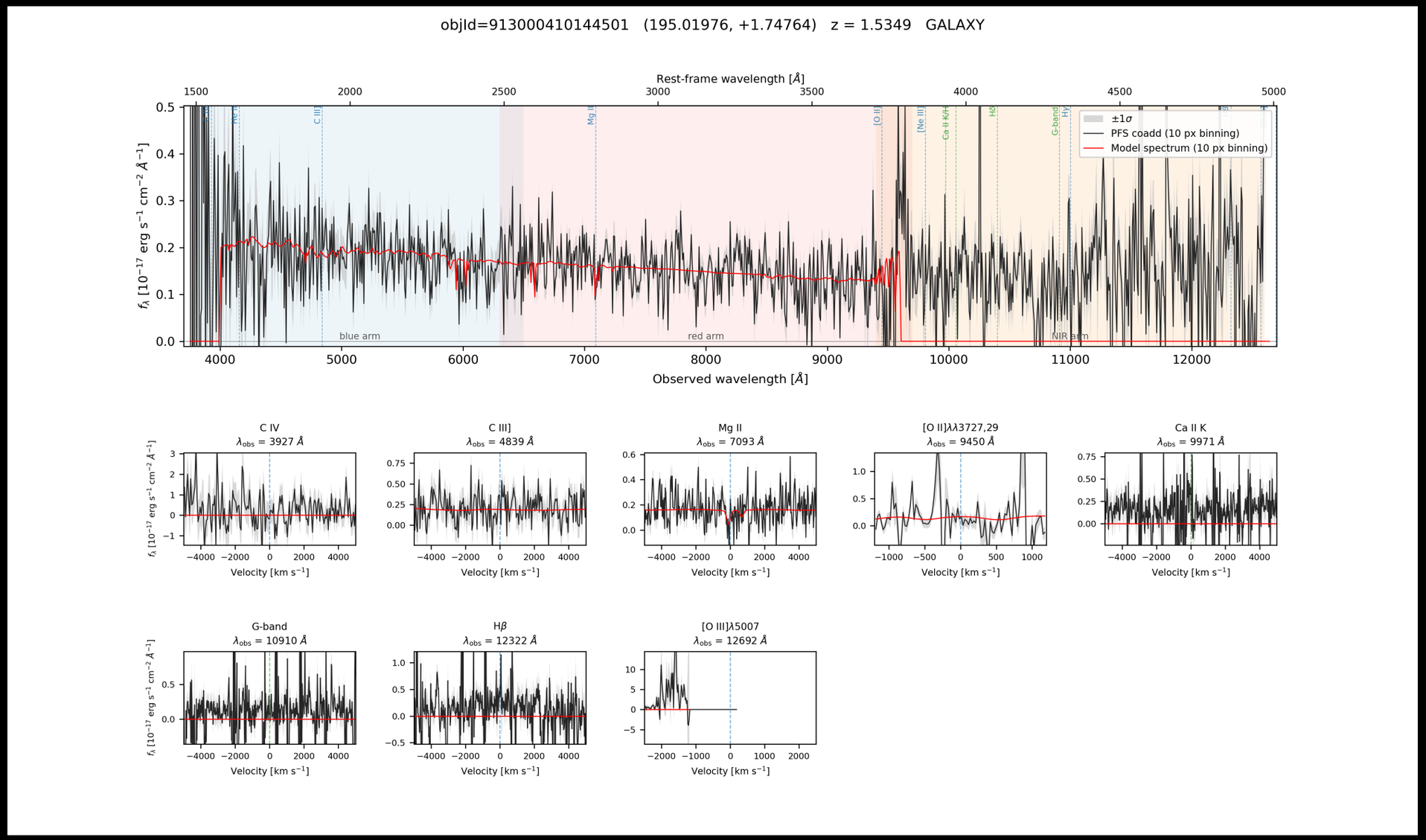

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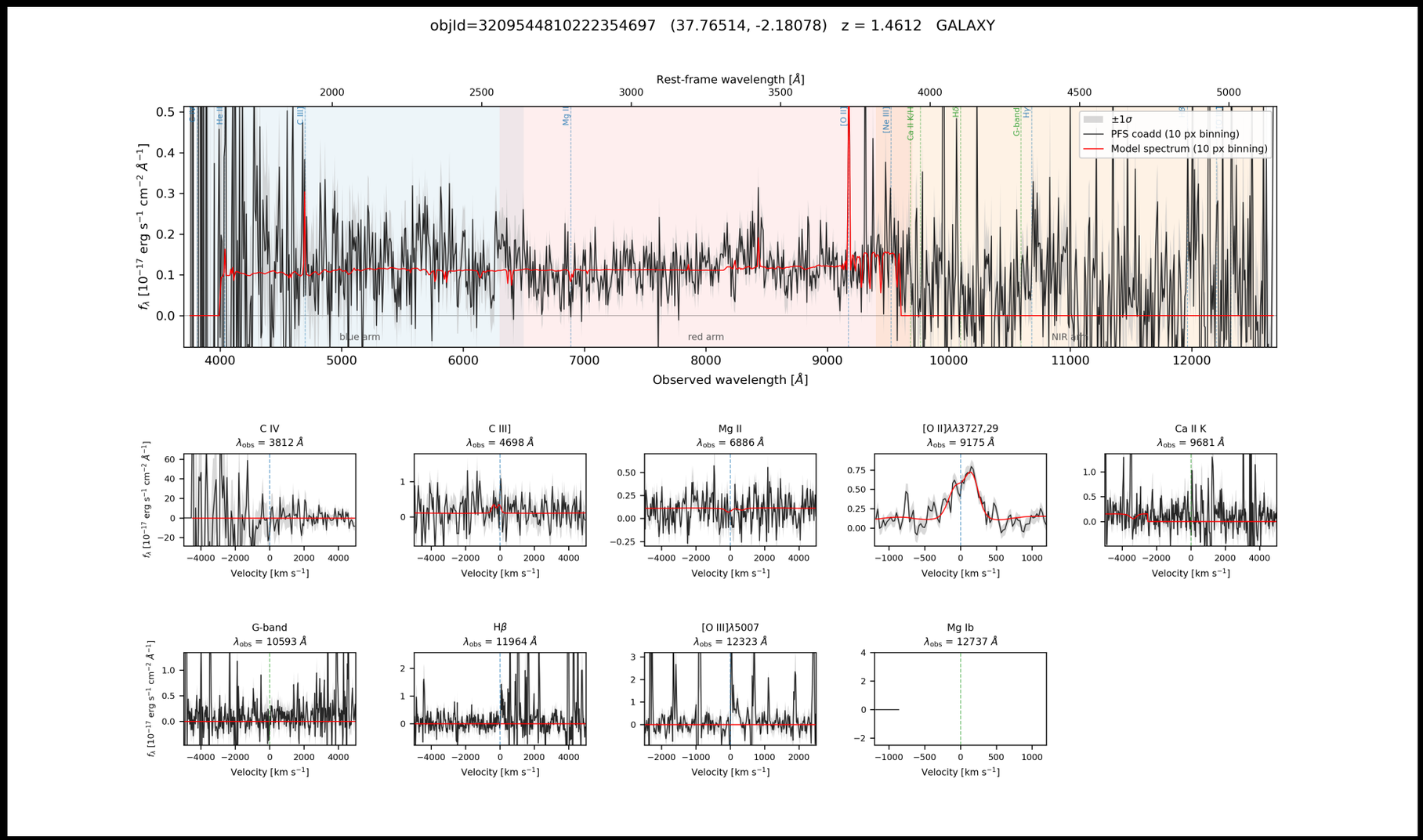

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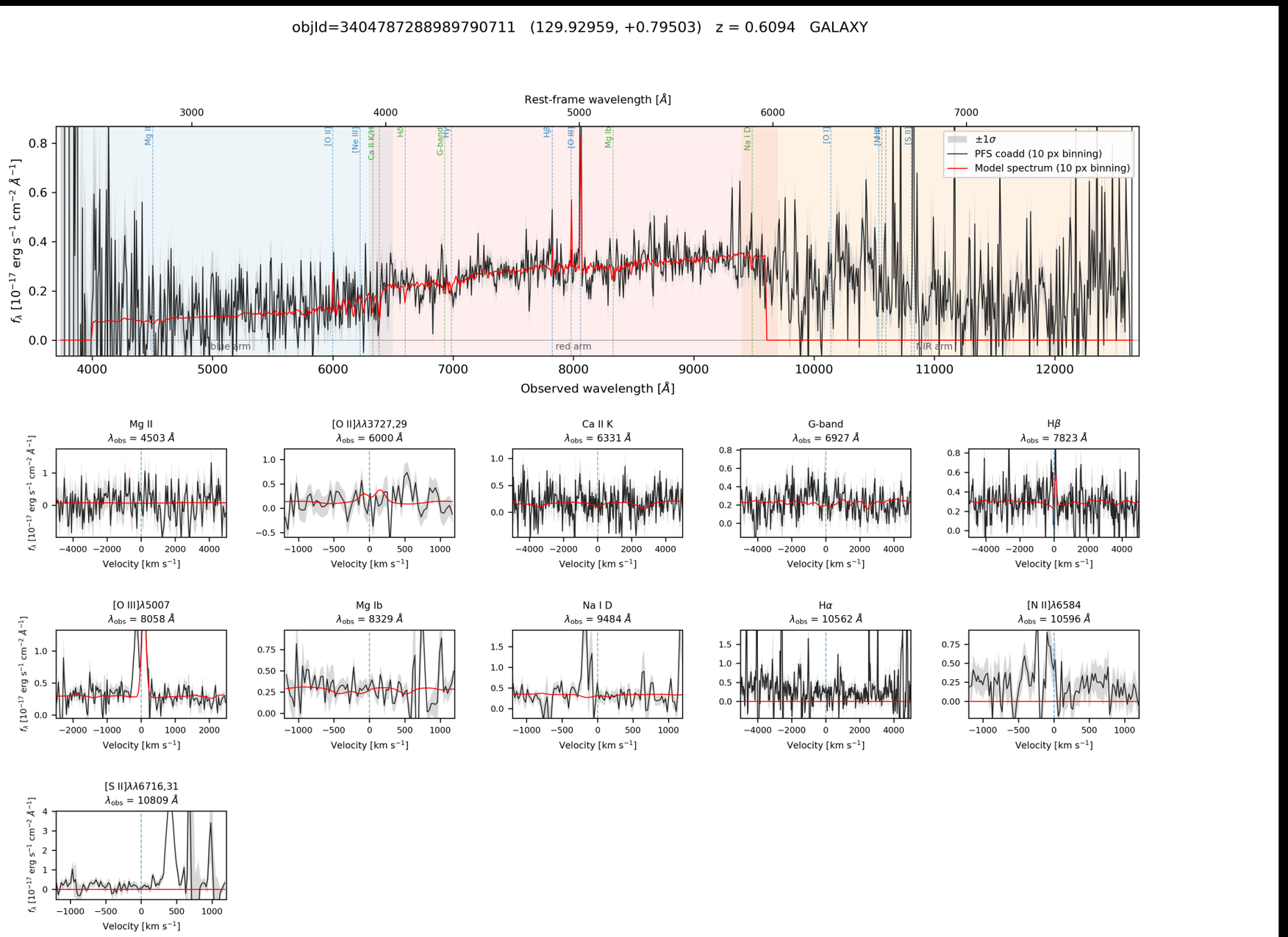

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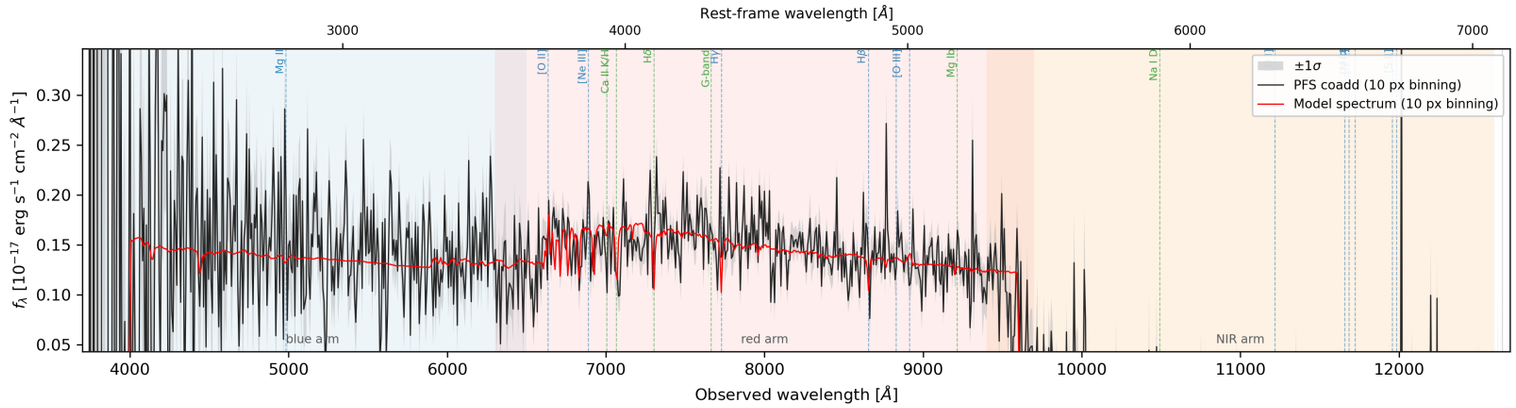
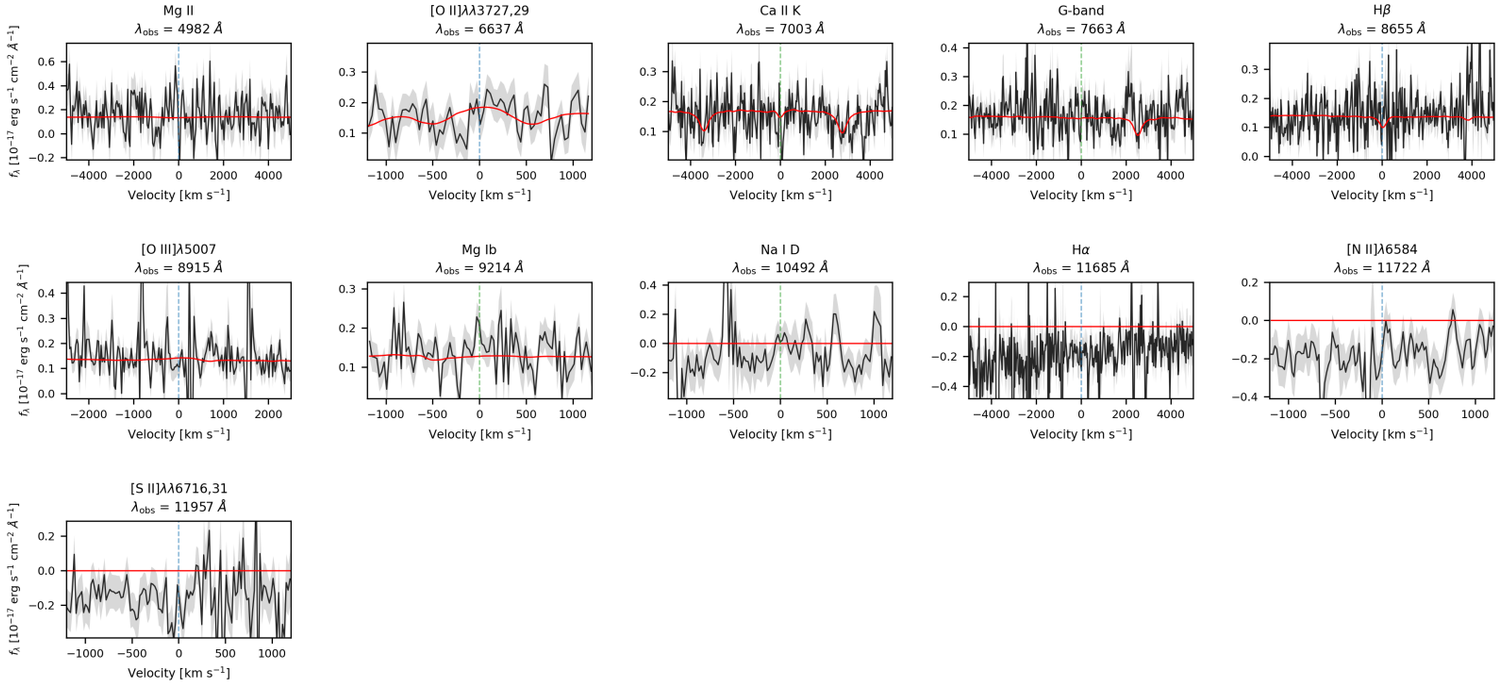
objId=3454243322007159542 (336.84621, +0.54943) z = 0.7804 GALAXY





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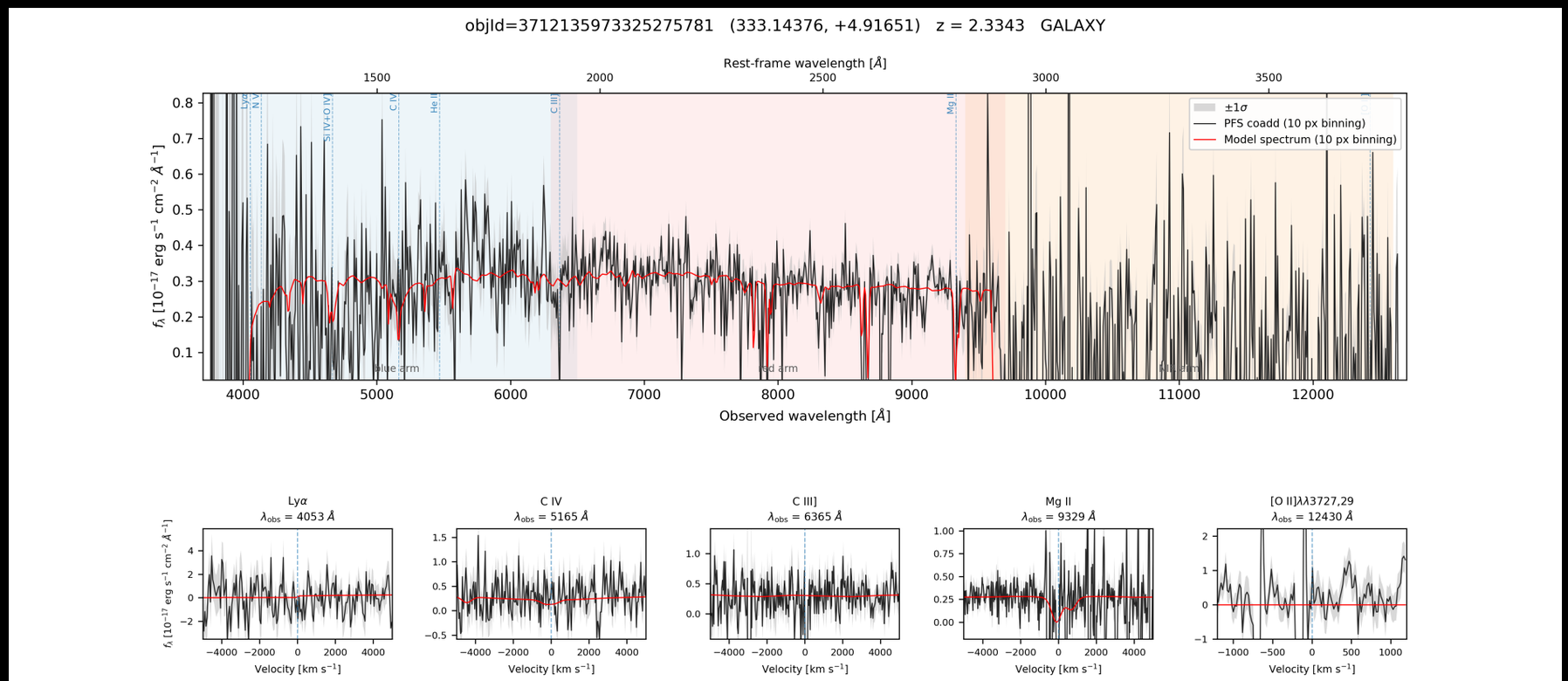

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